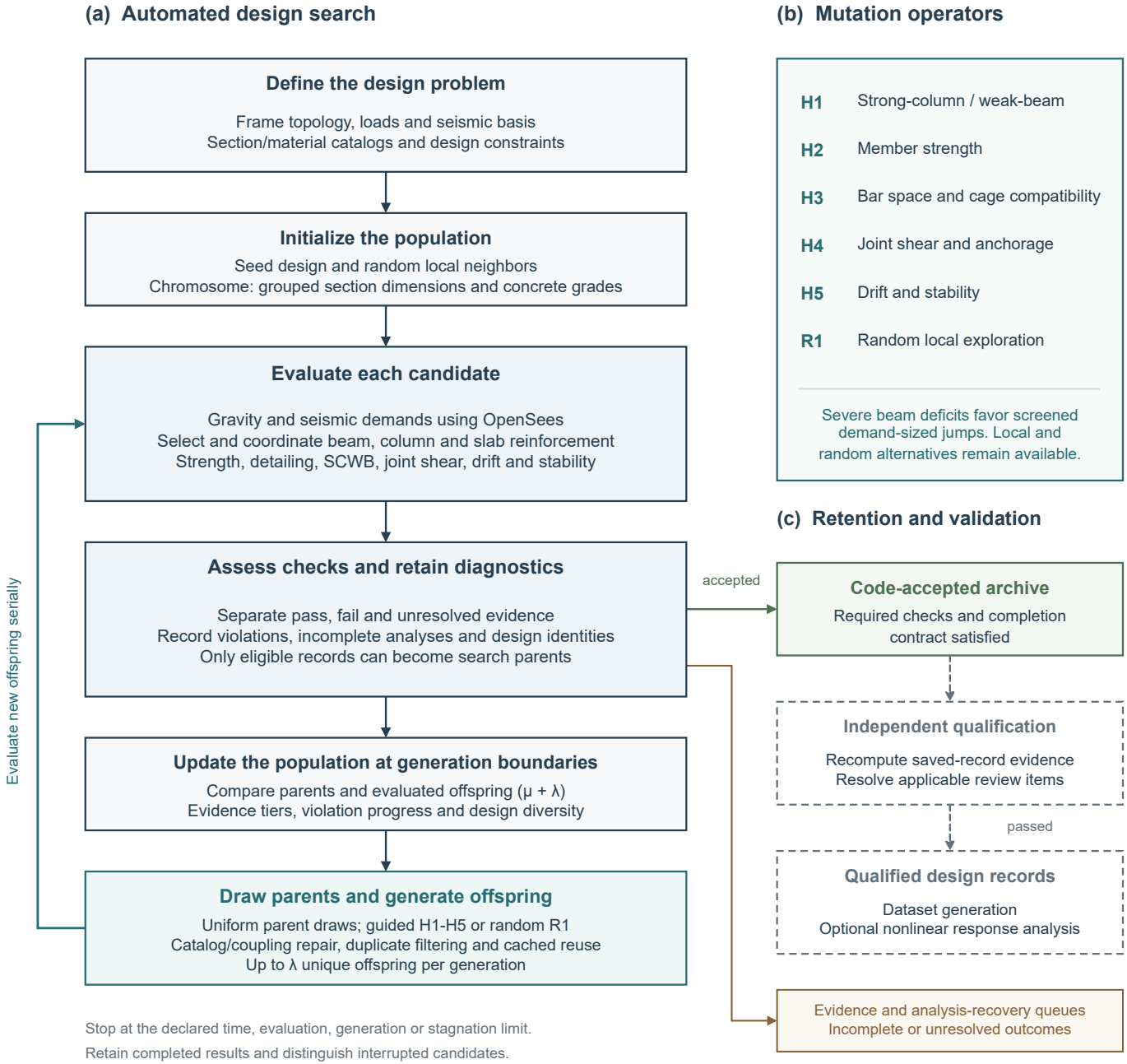

(a) Automated design search
Define the design problem
Frame topology, loads and seismic basis
Section/material catalogs and design constraints
Initialize the population
Seed design and random local neighbors
Chromosome: grouped section dimensions and concrete grades
Evaluate each candidate
Gravity and seismic demands using OpenSees
Select and coordinate beam, column and slab reinforcement
Strength, detailing, SCWB, joint shear, drift and stability
Assess checks and retain diagnostics
Separate pass, fail and unresolved evidence
Record violations, incomplete analyses and design identities
Only eligible records can become search parents
Update the population at generation boundaries
Compare parents and evaluated offspring (μ + λ)
Evidence tiers, violation progress and design diversity
Draw parents and generate offspring
Uniform parent draws; guided H1-H5 or random R1
Catalog/coupling repair, duplicate filtering and cached reuse
Up to λ unique offspring per generation
Evaluate new offspring serially
Stop at the declared time, evaluation, generation or stagnation limit.
Retain completed results and distinguish interrupted candidates.
(b) Mutation operators
H1 Strong-column / weak-beam
H2 Member strength
H3 Bar space and cage compatibility
H4 Joint shear and anchorage
H5 Drift and stability
R1 Random local exploration
Severe beam deficits favor screened demand-sized jumps. Local and random alternatives remain available.
(c) Retention and validation
accepted
Code-accepted archive
Required checks and completion contract satisfied
Independent qualification
Recompute saved-record evidence
Resolve applicable review items
passed
Qualified design records
Dataset generation
Optional nonlinear response analysis
Evidence and analysis-recovery queues
Incomplete or unresolved outcomes